

Calvin Grabowski

Technical Skills

Languages: C++, Python, C, SQL Java, JavaScript

Frameworks & Libraries: React, Next.js, FastAPI, Node.js, Matplotlib, Scikit-learn

Tools & Systems: GitHub, AWS (S3, Lambda, Rekognition), Linux, Postman

Education

Oregon State University

Corvallis, OR

Honors B.S. in Applied AI Computer Science (AMP)

Sept 2024 – Jun 2027

GPA: 3.98/4.0 (Honor Roll)

Coursework: Algorithms (CS 514), Operating Systems (CS 574), Linear Algebra, Databases, Software Engineering

Projects

EdgeGuard Hybrid Intelligence | *Python, AWS (S3, Lambda, Rekognition), OpenCV*

Jan - Mar 2026

- Implemented background subtraction and scoring methods for frame extraction, effectively reducing environmental noise by 90% during live video stream
- Designed frame selection algorithm (central motion, low blur) to improve classification accuracy
- Reduced video data 98% (59/60 FPS) via frame selection, up to \$0.059/sec saved in cloud costs
- Integrated AWS S3, Lambda, and Rekognition to process detections and trigger automated actions
- Demo Video:** <https://youtu.be/Z4a7dnCW41Y>

Small Shell Integration | *C, Linux*

Feb – Mar 2026

- Engineered a custom shell in C with a robust parsing engine for I/O redirection, signal handling, and background process management using `fork()`, `exec()`, and `waitpid()`

Spotify Playlist Analyzer (AI/ML) | *Python, FastAPI, Spotipy, RapidAPI, Matplotlib*

Sept – Dec 2025

- Architected a full-stack web application with a FastAPI backend to classify playlist moods using a custom-trained machine learning model
- Fetches tracks via Spotipy and audio features via RapidAPI for model input
- Visualized mood distributions with matplotlib and deployed FastAPI backend for ML inference

Experience

Front-End Developer Intern

(Virtual) Seattle, WA

Global Radio Outreach

Aug – Sept 2025

- Engineered reusable UI components using React and Next.js within a 20-person Agile team
- Optimized frontend component architecture using HTML, CSS, and JavaScript to improve overall UI responsiveness and usability

Volunteer Software Engineer

Colorado Springs, CO

LightSys Technological Services

Mar 2025 (Spring Break)

- Built an Android app with intuitive UI/UX for endangered Evenki language learners
- Architected multimedia support handling audio and image I/O natively on the device using Kotlin
- Published:** <https://play.google.com/store/apps/details?id=org.lightsys.learnevenki>

Leadership & Clubs

AI Club: Collaborated on a 10-week AI project focused on playlist mood. Kaggle competitions biweekly classification

Running Club: Trains 5 days per week; ran a 65-mile ultra run and a 195-mile relay (2/350 teams) and 4:29 mile

Summer Camp Counselor: Led 8-16 campers at Open Window School for two years (Bellevue, WA)

Tech Admin: Learned crimping thernet, changing internet access, watching surveillance, and upkeeping the website